

# Apache Tomcat Setup Guide

## 1. Overview

This document provides a structured guide to installing and configuring Apache Tomcat.

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## 2. Installation Methods

### 2.1 Install via APT (Recommended for Quick Setup)

**Tomcat 9 (javax):**

```
sudo apt install tomcat9 -y
```

**Tomcat 10 (jakarta):**

```
sudo apt install tomcat10 -y
```

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### 2.2 Manual Installation (Tomcat 9)

*Step 1: Download Package*

```
wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.80/bin/apache-tomcat-9.0.80.tar.gz
```

*Step 2: Extract Archive*

```
tar -xvzf apache-tomcat-9.0.80.tar.gz
```

*Step 3: Set Ownership*

```
sudo chown -R ubuntu:ubuntu apache-tomcat-9.0.80
```

*Step 4: Start Tomcat Server*

```
cd apache-tomcat-9.0.80/bin  
./startup.sh
```

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## 3. Configuration

### 3.1 Configure User Roles and Access

**File Location:**

```
/home/ubuntu/apache-tomcat-9.0.80/conf/tomcat-users.xml
```

**Edit Command:**

```
vi /home/ubuntu/apache-tomcat-9.0.80/conf/tomcat-users.xml
```

**Add the following roles and user inside <tomcat-users>:**

```
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<role rolename="admin-gui"/>

<user username="admin" password="admin123"
      roles="manager-gui,manager-script,admin-gui"/>
```

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## 3.2 Enable Remote Access to Manager Application

**File Location:**

```
/home/ubuntu/apache-tomcat-9.0.80/webapps/manager/META-INF/context.xml
```

**Modify the following property:**

From:

```
allow="127\.\.\d+\.\.\d+\.\.\d+|:::1"
```

To:

```
allow=".*"
```

 Note: Allowing all IPs (".\*") is not recommended for production environments.

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## 3.3 Change Default Port

**File Location:**

```
/home/ubuntu/apache-tomcat-9.0.80/conf/server.xml
```

**Update Connector Port:**

From:

```
<Connector port="8080" ... />
```

To:

```
<Connector port="9090" ... />
```

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## 4. Restart Tomcat

After making configuration changes, restart the server:

```
./shutdown.sh  
./startup.sh
```